Probing datasets: 8 RNAs, all probed in vitro

Dataset ID	RNA	Class	N (nt)	Reagent	R (nt)	|E|	Source
TRP4P6_DMS_0007	P4-P6 domain, Tetrahymena ribozyme	structured	202	DMS	11–51	22	RMDB
RNASEP_DMS_0000	RNase P catalytic domain	structured	265	DMS	4–53	22	RMDB
HIV3PR_DMS_0001	HIV-1 3' UTR	viral UTR	261	DMS	34–83	20	RMDB
ROTA_RFS11_DMS	Rotavirus segment 11 (full)	viral mRNA	667	DMS	6–55	19	RASP2
HOXA9_STD_0000	Hox A9 mRNA 5' UTR	mRNA UTR	1266	1M7	1–100	100	RMDB
HOXA5_STD_0000	Hox A5 mRNA 5' UTR	mRNA UTR	237	DMS	1–50	31	RMDB
CSDE1_DMS_0000	Csde1 mRNA, nt 190–386	mRNA	197	DMS	1–50	39	RMDB
YXJA_EQR220_BZCN	B. subtilis yxjA riboswitch leader	5' leader	220	BzCN	4–53	50	RASP2

High reactivity → **nucleotide unpaired**

DMS reads A, C · **SHAPE** (1M7, BzCN) reads A, C, G, U

R = first ~50 nt with data at the 5' end

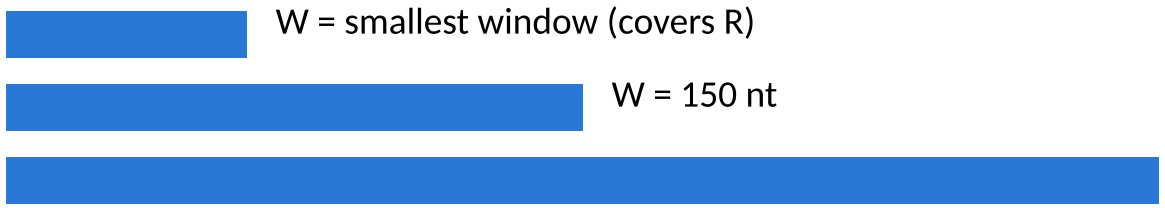
E = nucleotides of R that are scored (|E| = 19–100)

The test: does a window change the prediction?

Full RNA, length N



Windows, anchored at the 5' end



Fold each window: **RNAfold · CONTRAfold · MXfold2**

u(W) = probability that each nucleotide is unpaired

Same E for every window and model

A. Agreement with data

ρ(W) = Spearman(u(W), reactivity) over E

Δρ(W) = ρ(N) – ρ(W)

Δρ > 0: window agrees **worse** than full fold

B. Change of the model itself

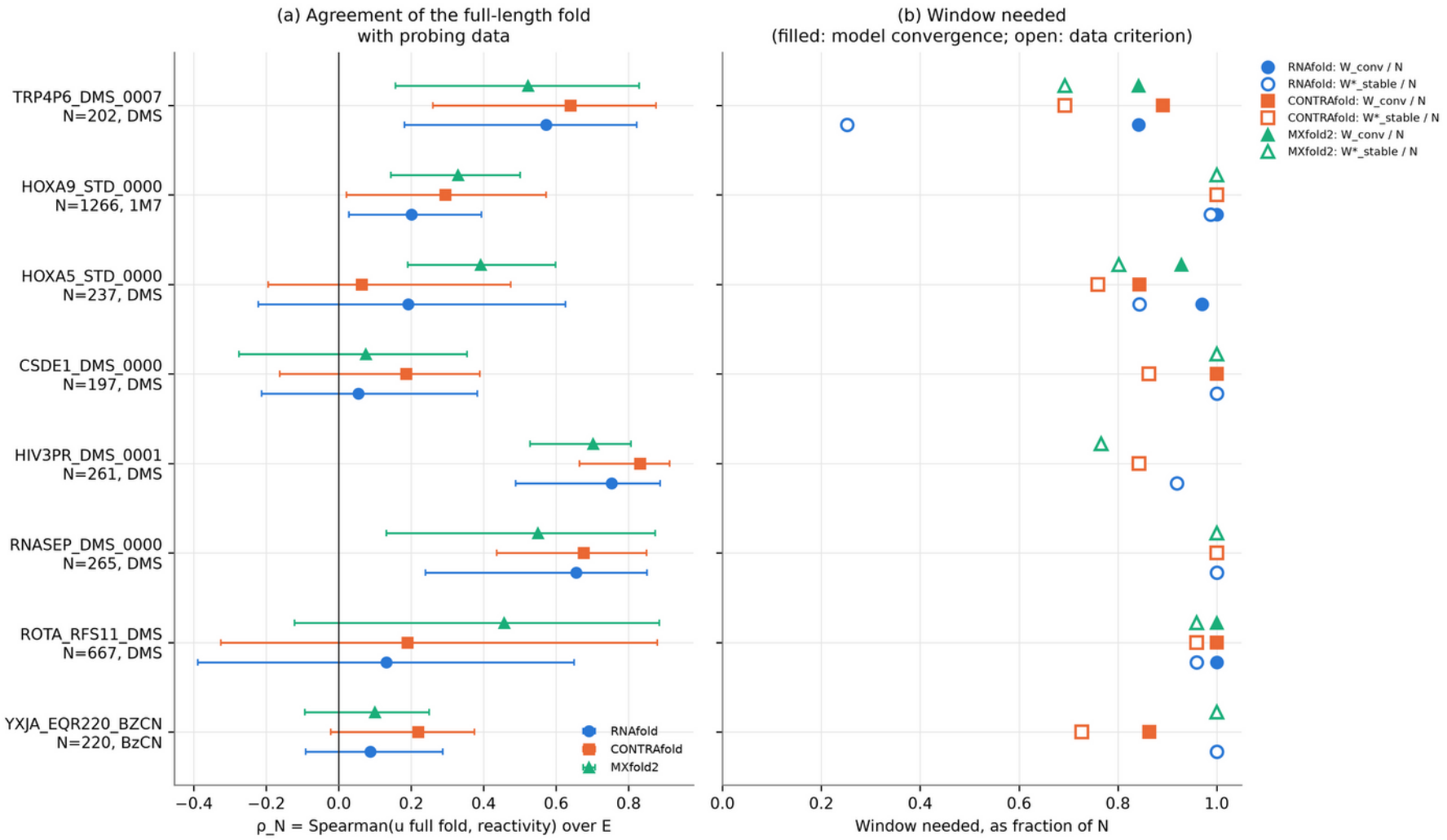
max over E of |u(W) − u(N)|

converged window: from here on ≤ 0.1

no experimental data used

95% CI: paired block bootstrap over positions
(2000 resamples, 10-nt blocks)

Full-length agreement, and the window each RNA needs



(a) Bars: 95% block-bootstrap CI. (b) W_conv: smallest W from which max |Δu| over E stays ≤ 0.1. W*_stable: smallest W from which Δρ ≤ ε for all larger W. 1.0 = the full length is needed.

(a) ρ(N): full fold vs data

bar = 95% CI

structured: 0.5–0.8

mRNA / UTRs: 0.05–0.46

(b) window needed / N

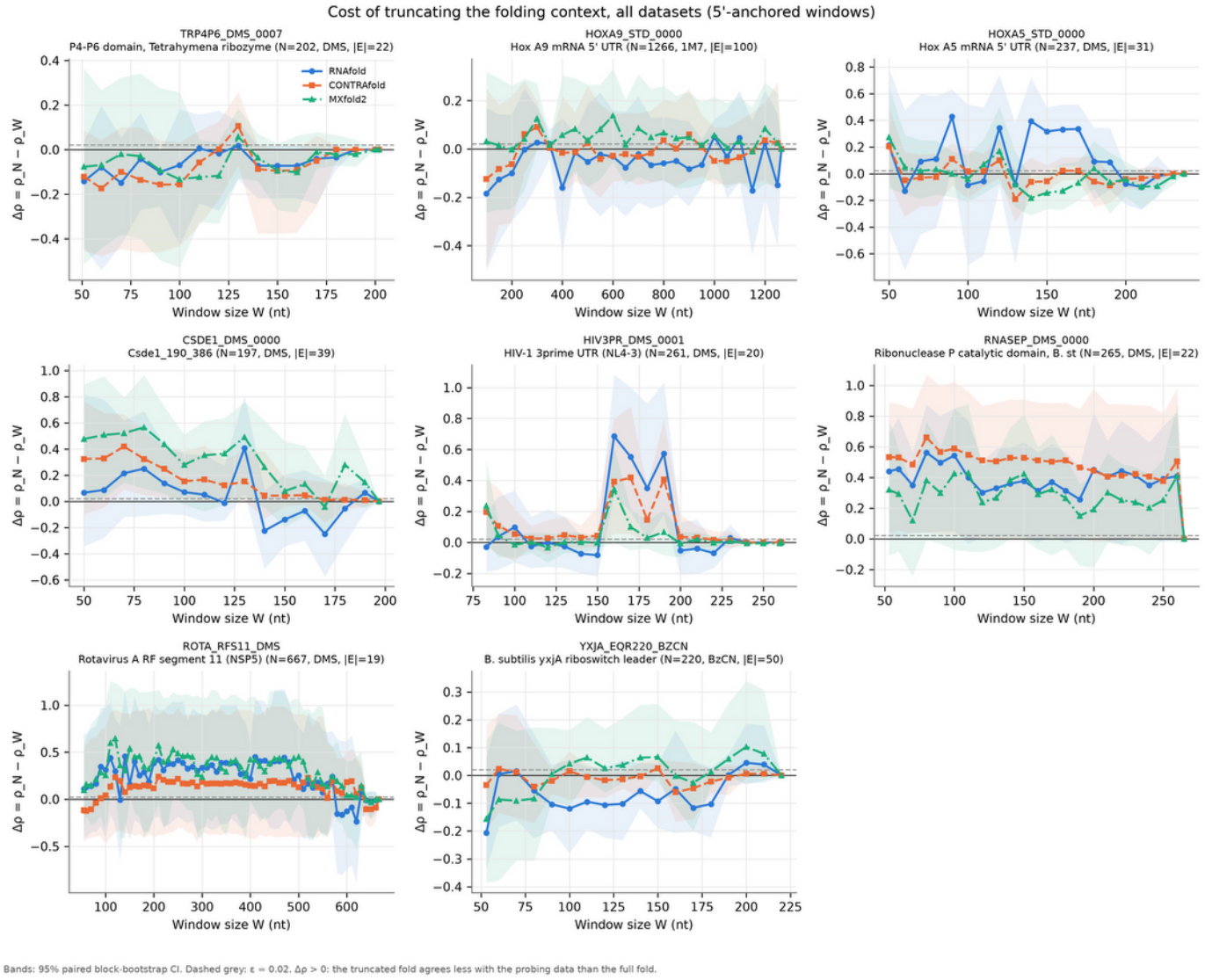
filled: model converged

open: data criterion met

almost all ≥ 0.7 N

except RNAfold on P4-P6 (0.25 N)

Cost of truncation depends on the RNA



How to read

x: window size W (nt) · y: Δρ(W) = ρ(N) − ρ(W)

above 0: window agrees **worse** with data

bands: 95% CI · dashed: ε = 0.02

RNase P: Δρ ≈ 0.3–0.6 at every W < N

HIV 3' UTR: fine at 100 nt, jump at 160–190 nt

Csde1: up to 0.5 for W ≤ 130 nt

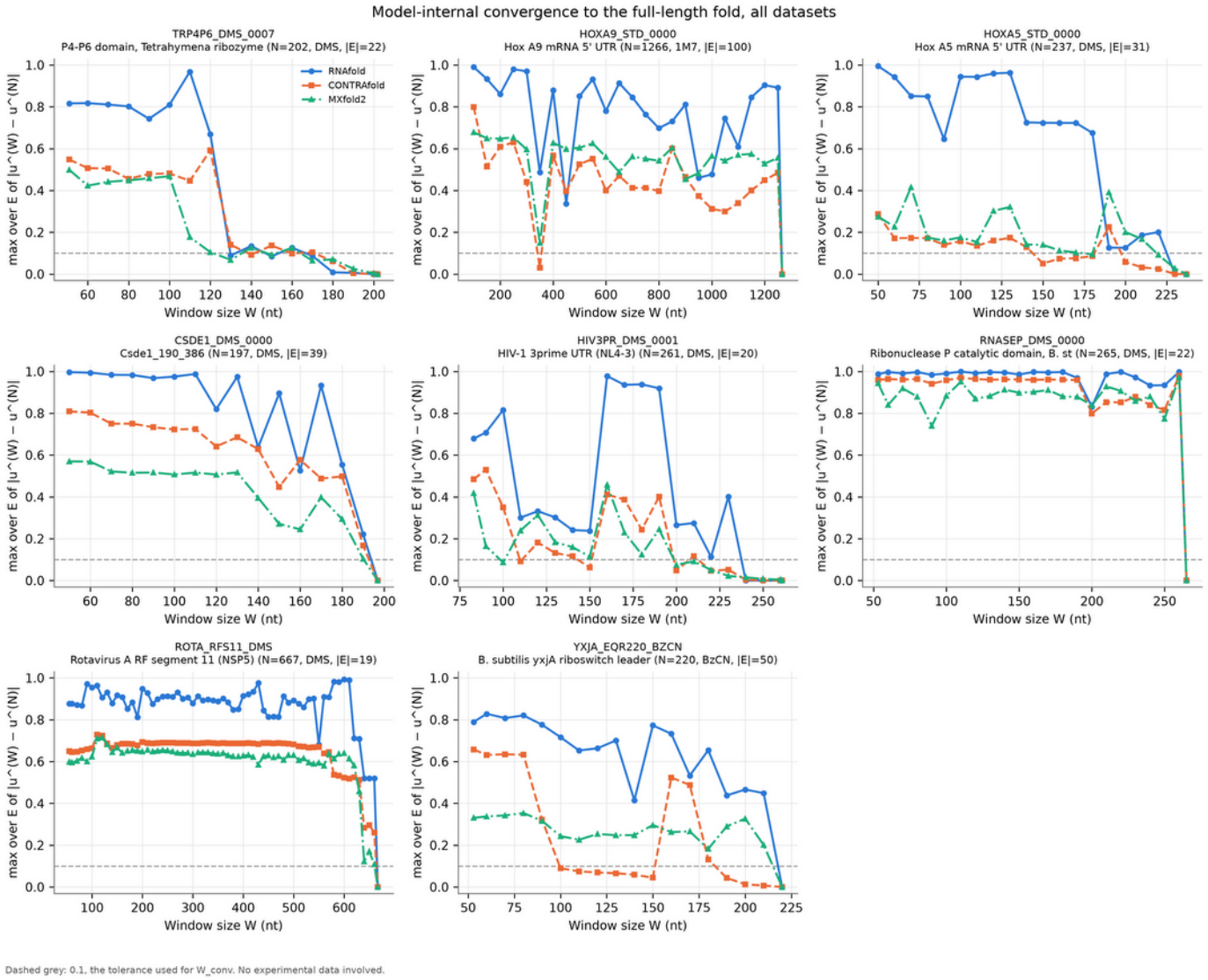
P4-P6, Hox A5/A9, yxjA: no clear cost

Rotavirus: ~0.2–0.4, very wide CIs

Across 8 RNAs: **no significant mean cost**

Wilcoxon p ≥ 0.25 at W = 100 and 150 nt

The prediction converges only near full length



How to read

x: window size W (nt)

y: max over E of |u(W) − u(N)|

dashed: 0.1 (converged) · no data used

Converged only at ≥ 0.77 N, every RNA and model

RNase P, Hox A9, Csde1, rotavirus: only at W = N

P4-P6, Hox A5, HIV: from 0.77–0.85 N

RNAfold moves most: often > 0.8

A short window gives a **different fold** even where Δρ is not significant

Takeaways: which window to use?

Fixed window



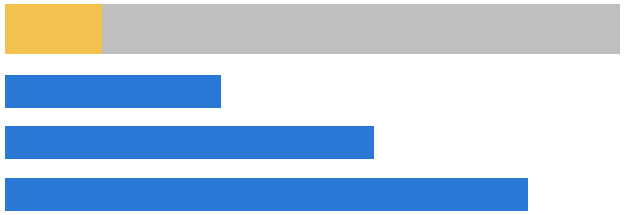
avoid: cuts helices, RNA-specific

Full length



default (done here up to 1.3 kb)

Adaptive window



if the RNA is too long

No fixed window is safe: the needed window depends on each RNA

Bigger ≠ safer: a window can cut a helix (HIV, 160–190 nt)

Policy: fold the **full length**; if too long, grow W until

max $|u(W) - u(N)| \leq 0.1$ over the region, **per RNA**, no data needed

Caveats: 8 RNAs, in vitro · 5' regions only ($|E| = 19–100$) · wide CIs
mRNA UTRs agree weakly with data even at full length